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TECHNICAL PAPER



Disassembly sequence planning of waste auto parts

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ABSTRACT

The disassembly of used products is a critical procedure in remanufacturing, and different disassembly strategies are often obtained from different perspectives. To describe the disassembly process more accurately, the uncertainty of the information in the disassembly process should be considered. Therefore, random variables are introduced for disassembly time, cost, and effort. Based on the extended stochastic Petri net modeling method and stochastic programming theory, a stochastic optimization algorithm combined with artificial intelligence technology and a multiobjective genetic algorithm are designed, and a multiobjective optimization model for the disassembly sequence of used car parts under uncertain conditions is successfully constructed. This model considers the viewpoint of the decision maker. Moreover, the Monte Carlo method is applied to solve the multiobjective optimization model, and the validity and practicability of the model are verified by an example of an automotive transmission.

Implications: With the rapid development of the economy and the shortening of the product life cycle, the rate of product renewal is getting faster and faster, which also leads to the production of a large number of waste products. According to the forecast of the relevant departments, it is estimated that, in 2020, there will be about 35 million used televisions, 15 million used refrigerators, 13 million used washing machines, 12 million used air conditioners, 57 million used computers, and 8.3 million scrapped cars. Waste products contain a lot of renewable resources. If they cannot be effectively recycled, it will be a great waste of resources, and unreasonable disposal of waste products may have a negative impact on the environment. Therefore, due to environmental pressure and economic drive, product recycling and remanufacturing activities have caused widespread concern in society. Disassembly is defined as the operation or activity of disassembling an assembly such as a product, assembly, or component, and is the result of multiple removal operations of the product. It is a prerequisite for the efficient recycling of products and the first link in remanufacturing, that is, disassembly as a new production activity, which can provide raw materials for the smooth progress of the remanufacturing production plan, namely, old rough or used parts. Efficient dismantling not only saves natural resources and energy, but also effectively reduces environmental pollution. It is also an important guarantee for promoting the healthy development of the circular economy and achieving sustainable industrial development. The length of time required for the dismantling process, the level of costs, and the amount of profits obtained will directly affect the economic benefits of the recycling of end-of-life products. Therefore, the evaluation and optimization of the dismantling process of waste products have become one of the current hot issues. The research on the dismantling of waste products is conducive to speeding up the recycling process of waste, to a greater extent, the rapid and full recovery of resources, and to a certain extent, it will provide value basis and theoretical significance for subsequent research.

PAPER HISTORY

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Introduction

Disassembly planning is a decision-making process of disassembly sequence planning based on the optimal objective function. From the whole process of product design, Kroll and Carver (1999) studied the difficulty of product disassembly and put forward the evaluation method of product disassembly, including disassembly type, disassembly accessibility, disassembly strength, and other indicators, which laid a theoretical foundation for the generation of other evaluation indicators. Tian

et al. (2012) emphasized the difference between disassembly and dismantlement in the process of disassembly of waste products, defined the concept of depth and degree of disassembly, and put forward the evaluation method and planning model based on the probability of product disassembly.

Homem de Mello and Sanderson (1991) proposed an algorithm for generating the sequence of all mechanical devices for a given product by using the component relationship model. At this time, the disassembly

problem was decomposed into different disassembly subproblems, and the disassembly task was transformed into an inverse process of the assembly task. Zussman and Zhou (2000) proposed a disassembly planning method based on Petri nets model, and proposed three kinds of planning and execution algorithms (global fixing, finite adaptive, and adaptive) for disassembly process, which realized the optimization of product disassembly process and provided theoretical guidance for product remanufacturing process. In order to maximize the disassembly profit, Lambert (Lambert 2006a, 2006b, 2007) used dynamic planning to optimize the disassembly sequence, and proposed an improved two-commodity network flow method, which reduces the number of integer variables and is suitable for network graph model with disassembly priority. Then, by solving the binary integer linear programming problem, he completely avoided the use of integer variables, and proved the effectiveness and accuracy of the method. Li, Khoo, and Tor (2005) proposed an object-oriented modeling method for generating disassembly sequence from disassembly constraint graph, and applied genetic algorithm to solve the optimization problem of feasible disassembly sequence. Adenso-Díaz, García -arbajal, and Gupta (2008) proposed a stochastic adaptive greedy search algorithm and a heuristic algorithm for path reengineering, aiming at the lowest cost and shortest time of disassembly, which is the first key link in the product recycling process, and verified the effectiveness of the algorithm, which can be used to solve the bi-objective disassembly path problem. Aguinaga et al. (2008) analyzed the similarities and differences between the disassembly problem and the general path planning problem, and proposed an improved fast growing random tree algorithm to solve the problem that the extraction trajectory of parts may not be predicted in the disassembly process.

It is noteworthy that the above research on disassembly of mechanical devices belongs to the category of certainty. In the actual disassembly process of mechanical devices, because the products to be disassembled are used in different environments for different periods of time, human or objective uncertainties will be encountered in the disassembly process, such as the quality of disassembly personnel, the degree of damage to objects to be disassembled, disassembly tools, disassembly environment, and so on, which cannot be ignored in the process of product disassembly.

Tian, Zhou, and Chu (2013) proposed a disassembly multiobjective stochastic optimization model that combines stochastic simulation, genetic algorithm, and neural network, and analyzed its convergence and other mathematical properties. They also use the

stochastic weight method to transform the multiobjective optimization function to the single-objective optimization function, in order to obtain the compromise or Pareto solution of the multiobjective stochastic optimization of product disassembly. In view of the increasing environmental and economic pressures in the process of disassembly, and the influence of human interference on disassembly, Tang, Zhou, and Gao (2006) developed a fuzzy attribution Petri nets (FAPN) model to mathematically express the disassembly uncertainty caused by a large number of human interventions, and proposed an algorithm on this basis for optimal disassembly planning, which is more suitable for actual industrial processes. Gao, Zhou, and Tang (2004) proposed a fuzzy reasoning Petri nets (FRPN) model to represent the relevant decision-making rules in the process of disassembly in order to solve the problem of great uncertainty in the products used, and used the fuzzy reasoning algorithm based on FRPN model to automatically and quickly make the most decision considering multicriteria disassembly rules. Grochowski and Tang (2009) used a machine learning method based on disassembly Petri nets (DPN) and hybrid Bayesian networks (HBN), which modeled the disassembly process by checking the probabilistic relationship between different aspects of the disassembly process and predicted the results of each disassembly process. The above researchers have mainly discussed and studied the disassembly evaluation, disassembly sequence planning and disassembly solutions under uncertain conditions, which provided some theoretical support for the writing of this paper.

Disassembly planning consists of two major steps: disassembly model construction and disassembly sequence planning (DSP) (Tian et al. 2019). Although some researchers have paid attention to the problem of random planning of product disassembly, on the basis of this, this dissertation is mainly based on the construction of a disassembly information model based on advanced Petri nets; according to the actual disassembly decision requirements, a typical mathematical planning model is established; design combining artificial intelligence technology with stochastic optimization algorithm and multiobjective genetic algorithm (Hou et al. 2017); analysis of disassembly modeling and comprehensive optimization of waste products in uncertain environments; and analysis and application verification of related theories and methods.

The rest of the article is divided into the following aspects. The second part describes the establishment of the multiobjective disassembly optimization model. The third part is the Monte Carlo algorithm design. The fourth part mainly describes the analysis of specific cases. The fifth part describes the solution of the

established model. Finally, it summarizes and describes the deficiencies of the research.

Multiobjective stochastic disassembly optimization model

In order to present the different requirements of decision makers for the disassembly process of waste products in actual disassembly activities, the minimum expected value model, the maximum disassembly degree model, and the minimum optimization model with disassembly time, disassembly fee, and disassembly energy as multi objectives are established in this section. In order to simplify the model and express it conveniently, the fee of disassembly, the time of disassembly, and the energy of disassembly are collectively referred to as the fee of disassembly—the fee of disassembly time, the fee of disassembly fee, and the fee of disassembly energy. It should be noted that in the process of solving the multiobjective stochastic optimization model (Kang et al. 2017), it is impossible to obtain the theoretical optimal solution, so it is necessary to apply the Monte Carlo method to solve the Pareto optimal solutions of the model.

In practical applications, there are many modeling methods for the dismantling of mechanical devices. Directed graph models are generally expressed using the binary variable $G(V, E)$, and undirected graphs are also generally expressed using the binary variable $G(V, E)$. The AND or graph model is generally expressed by the binary variable $G(W, D)$, which is a modeling method that can effectively express the logical relationship between elements and the internal structure information of the model, and the Petri net model is a highly efficient, A modeling method for quantitatively and graphically processing discrete events (Tian et al. 2018).

The Petri net model is a modeling method that can process discrete events efficiently, quantitatively, and graphically. The Petri net model is mainly composed of a place (P), a transition (T), a token, and a directed arc. It can vividly represent the direction of information flow and qualitatively calculate the resources occupied by each link. The construction of a disassembly information model is the basis for product disassembly optimization. The construction of product disassembly information model under uncertain environment is a difficult point of the article. Considering that Petri nets can simply and accurately represent the advantages of concurrency, conflict, uncertainty, and dynamic process description of dismantling systems, this article intends to use extended random construction of product disassembly information model in uncertain environment based on the Petri net theory.

This article uses the Petri net six-tuple variable PN to describe the dismantling Petri net model of the mechanical

device, where $PN = (P, T, A, S, W, R, M_0)$, and now describes the parameters of the six-tuple variable PN as follows.

(1) $P = \{P_1, P_2, \dots, P_m\}$ is a collection of finite warehouses representing m parts, and the element P_i contained in it is the i -th part of the mechanical device to be dismantled, and i satisfies the relationship $1 \leq i \leq m$, where m is the total number of parts of the product.

(2) $T = \{T_1, T_2, \dots, T_n\}$ is a finite transition set representing n disassembly operation steps of m parts, T_j represents a disassembly operation between two parts, and j satisfies the relationship $1 \leq j \leq n$, where n is the sum of all disassembly steps of the product to be disassembled.

(3) A represents the directed arc between the place and the transition. The mathematical expression of matrix A can be expressed as follows

$$A = a_{[ij]} = \begin{cases} -w(i, j) & (\text{When the directed arc points from } p_i \text{ to } t_j) \\ w(i, j) & (\text{When the directed arc points from } t_j \text{ to } p_i) \\ 0, & (\text{other}) \end{cases}$$

(4) $S = \{S(P_1), S(P_2), S(P_3), \dots\}$ means a finite set of places in the constraint state of each part, and $S(P_i) = 1$ means that the constraint of part i has been lifted, $S(P_i) = 0$, the part i is still in a constrained state, which is as follows:

$$S_{ij} = \begin{cases} 0, & \text{Unconstrained} \\ 1, & \text{Bound} \end{cases}, 1 \leq i \leq m, 1 \leq j \leq n.$$

(5) $R = \{R(T_1), R(T_2), R(T_3), \dots\}$ is a finite set representing the changes that have occurred, obviously. If $T_i \in R$, it means that T_i is a change that has occurred; if, it means that T_i is a change that has not yet occurred.

(6) $M_0 = (a_1, a_2, \dots, a_i, \dots, a_m)^T$, where $i \in [1, m]$, $a_i \in \{0, 1\}$. That is, M_0 defines the initial identity of the Petri net. Each element in the vector M_0 represents the number of tokens contained in each library in the initial state, and the number of tokens in the initial state is at most 1.

(7) W : W is the weight function of each transition. What W reflects in the extended stochastic Petri net is the probability distribution subject to various changes (such as dismantling time, dismantling cost, dismantling energy). The value method of the weight function is as follows: first, determine the probability distribution that each change should obey, and then solve the optimal solution of the planning model according to the stochastic optimization theory.

Model assumption

1. Only when the parts are unconstrained can they be disassembled from the mechanical device, and the

variable $S(P_i)$ is used to indicate that the parts (place) P_i is unconstrained.

2. For such parts that are not necessary to be scattered sometimes in the actual process of disassembling mechanical devices, in order to prevent unnecessary wear and tear during disassembly and reduce disassembly time, they can be disassembled as a whole.
3. In the process of disassembling parts, the time consumed by changing disassembly tools is neglected.
4. All parts are rigid. In the process of disassembling parts, the disassembly direction is the action direction of two parts, but it is difficult to determine the disassembly direction of non-rigid parts (such as wires), so the disassembly direction of non-rigid parts is not considered in this article.
5. All parts are removable. In practice, the disassembly of all the parts connected to it may be affected by the damage of some parts. For convenience of research, it is assumed that all parts can be disassembled in this article.
6. Disassembly time, disassembly fee, and disassembly energy of each component of waste products are independent random variables.

Parameter definition

m = Total number of places, $m = 1, 2, 3, \dots$;
 n = Total number of transitions, $n = 1, 2, 3, \dots$;
 k = Feasible disassembly sequence number, $k = 1, 2, 3, \dots$;
 i = Removal operations corresponding to transition i , $i = 1, 2, 3, \dots, n$;
 j = Parts to be removed corresponding to place j , $j = 1, 2, 3, \dots, m$;
 t = Disassembly time constraints in the maximum disassembly degree model;
 s = Disassembly fee constraints in the maximum disassembly degree model;
 w = Disassembly energy constraints in the maximum disassembly degree model;
 p_1 = Disassembly time opportunity constraints in the minimum disassembly cost model;
 p_2 = Disassembly fee opportunity constraints in the minimum disassembly cost model;
 p_3 = Disassembly energy opportunity constraints in the minimum disassembly cost model;
 S = Vector of disassembly fee, the elements in the vector are the random disassembly fee needed to complete each disassembly operation;
 T = Vector of disassembly times, the elements in the vector are the random disassembly times needed to complete each disassembly operation;

W = Vector of disassembly energy, the elements in the vector are the random disassembly energy amounts needed to complete each disassembly operation;

M = Initial state identification column vector for the disassembly of a used product;

M_0 = Final state identification column vector for the disassembly of used product;

A = Incidence matrix, the incidence matrix in the stochastic extended Petri net model;

U = Vector of decision variables, if $U(i) = 1$, the i -th transition in the Petri net occurs; otherwise, it does not occur.

Minimum expected cost disassembly model

First, taking the minimum disassembly time, cost and energy as objective functions, the following multiobjective minimum expected disassembly fee model is constructed:

$$\min[E(\vec{T}\vec{u}_k), E(\vec{S}\vec{u}_k), E(\vec{W}\vec{u}_k)] \quad (1)$$

$$st. \begin{cases} \vec{A}\vec{u}_k = \vec{M} - \vec{M}_0 \\ \vec{u}_k(i) \in \{0, 1\}, i = 1, 2, \dots, n \\ \vec{M}(j) \in \{0, 1\}, j = 1, 2, \dots, m \\ |\vec{M}| = |\vec{M}_0| \end{cases}$$

Maximum disassembly degree model

In the actual disassembly process, the decision-maker may pay more attention to the possibility that a part can be disassembled within a specified time. The maximum cost disassembly model seeks the disassembly sequence u_k which is most likely to disassemble the target parts or completely disassemble the products within the specified disassembly time t , disassembly fee s , and disassembly energy w :

$$\max[P\{\vec{T}\vec{u}_k \leq t\}, P\{\vec{S}\vec{u}_k \leq s\}, P\{\vec{W}\vec{u}_k \leq w\}] \quad (2)$$

$$st. \begin{cases} \vec{A}\vec{u}_k = \vec{M} - \vec{M}_0 \\ \vec{u}_k(i) \in \{0, 1\}, i = 1, 2, \dots, n \\ \vec{M}(j) \in \{0, 1\}, j = 1, 2, \dots, m \\ |\vec{M}| = |\vec{M}_0| \end{cases}$$

Minimum cost disassembly model

In the actual disassembly process, if the decision-maker is willing to take risks, he can choose the sequence that may produce the minimum disassembly fee under a certain probability for disassembly operation:

$$\min[\vec{T}\vec{u}_k, \vec{S}\vec{u}_k, \vec{W}\vec{u}_k] \quad (3)$$

The constraints are as follows:

$$st. \begin{cases} \vec{A}\vec{u}_k = \vec{M} - \vec{M}_0 \\ \vec{u}_k(i) \in \{0, 1\}, i = 1, 2, \dots, n \\ \vec{M}(j) \in \{0, 1\}, j = 1, 2, \dots, m \\ |\vec{M}| = |\vec{M}_0| \\ P\{Tu_k \leq t\} \geq p_1 \\ P\{Su_k \leq s\} \geq p_2 \\ P\{Wu_k \leq w\} \geq p_3 \end{cases}.$$

Algorithm design

After the construction of the extended random Petri net model, the feasible disassembly sequence can be calculated according to the equation $\vec{A}\vec{U} = \vec{M} - \vec{M}_0$. The Monte Carlo method is to solve a problem. First, a probability model or a random process is established, so that its parameters or numerical characteristics are equal to the solution of the problem: then these parameters or digital characteristics are calculated by observation or sampling experiments on the model or process. Finally, give the approximated value. The Monte Carlo method is used to simulate the distributions of the total disassembly time, total fees, and energy for each disassembly sequence. The multiobjective Pareto solution sets of the minimum expected cost model, maximum disassembly degree model, and minimum disassembly cost model are calculated according to the simulation results. Assuming that there are k feasible disassembly paths, the specific algorithm is as follows:

Step 1: Initialize the random number, N , for the total number of simulation trials.

Step 2: Calculate the total disassembly time, disassembly fees, and disassembly energy of the corresponding probability distribution functions, $\varphi_i(\xi_1)$, $\varphi_i(\xi_2)$, and $\varphi_i(\xi_3)$, and the definition domain of the i -th feasible disassembly sequence, where $1 \leq i \leq k$ and $i \in Z$.

Step 3: Generate the random variables, ξ_1 , ξ_2 , and ξ_3 in the probability distribution function domain

Step 4: Repeat step 3 N times, and arrange the random variables, ξ_1, ξ_2 , and ξ_3 , in ascending order according to their numerical values and store them in the arrays, C_{i1} , C_{i2} , and C_{i3} , respectively

Step 5: Calculate the averages, μ_{i1} , μ_{i2} , and μ_{i3} , of the arrays, C_{i1} , C_{i2} , and C_{i3} , for the expected disassembly and the disassembly time, expected disassembly fees, and the desired disassembly energy of the i -th feasible disassembly sequence are obtained.

Step 6: m_{i1} , m_{i2} , and m_{i3} are the values of the elements that are smaller than t , s , and w in the C_{i1} , C_{i2} , and C_{i3} arrays; and $\frac{m_{i1}}{N}$, $\frac{m_{i2}}{N}$, $\frac{m_{i3}}{N}$ are the maximum time

disassembly degree, maximum fee disassembly degree, maximum energy disassembly degree of the i -th feasible disassembly sequence under the constraints t , s , and w

Step 7: Find the $N^{\bullet}p1$ -th, $N^{\bullet}p2$ -th, and $N^{\bullet}p3$ -th elements in the C_{i1} , C_{i2} , and C_{i3} arrays, respectively, which are the minimum disassembly times of the i -th feasible disassembly sequence under the constraints p_1 , p_2 , and p_3 , the minimum disassembly costs and the minimum disassembly energy.

After applying the above algorithm, the Pareto solution set of the disassembled sequence can be obtained by comparing the solution results of the k feasible disassembled sequences.

The basic elements and operating rules of the disassembled Petri net modeling theory and the basic Petri net modeling theory are roughly the same, but the disassembled Petri net has the following characteristics.

1) In the process of dismantling the product, there are dismantling in multiple directions, and there are complex inter-relationships and spatial constraints between parts. Therefore, when constructing the Petri net model, it is necessary to judge the connection relationship (AND/OR relationship) between the parts according to the actual design and structure of the product, and construct the disassembly priority matrix.

2) When generating the disassembled Petri net, in order to reflect the priority of disassembly and the spatial characteristics of the disassembled structure, it is necessary to insert appropriate places and changes in the disassembled Petri net to express this structural feature.

Case analysis

Case description

In this section, taking automotive transmission as an example, a comprehensive disassembly optimization model for an uncertain environment is established based on the Stochastic Extended Petri Net Theory (SEPNT) (Grochowski and Tang 2009; Tian et al. 2012), and its solution is analyzed. To facilitate the analysis, according to the actual situation, the profile of the automobile transmission device is drawn, as shown in Figure 1.

In the process of setting up the bolt disassembly experiment, this project mainly considers the influence of the following factors: (1) the disassembly state of the product, which is simulated by the tightening torque of the bolt; (2) the two types of disassembly tools are different. The disassembly tool disassembles the bolt and performs disassembly time statistics under different experimental conditions. The first is to use a ratchet wrench, which is represented by the

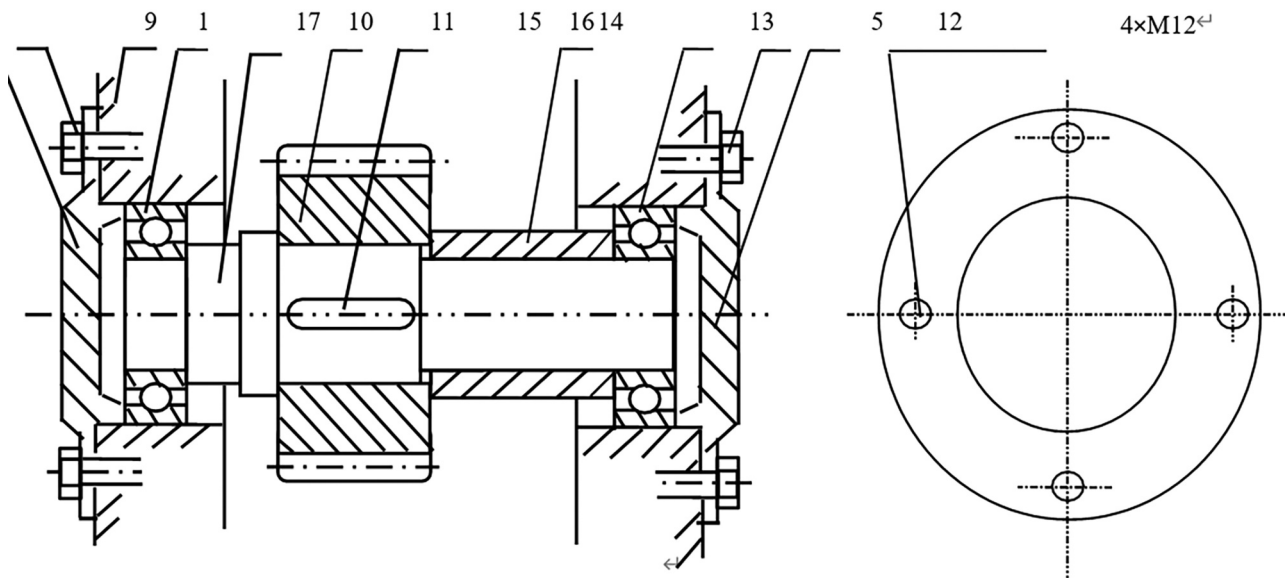


Figure 1. Typical automotive transmission profile. 1, 2, 3, 4 – Bolt I; 5, 6, 7, 8 – Bolt II; 9 – Left End Cap; 10 – Bearing I; 11 – Axis; 12 – Right End Cap; 13 – Bearing II; 14 – Bushing; 15 – Gear; 16 – Key; 17 – Cover.

tool code I; the second is to remove the plum wrench, which is represented by the tool code II; (3) the quality of the dismantling staff, the dismantling staff with different maximum pulling forces are used to perform the disassembly, and the largest Torque is measured with a dynamometer.

Note that the disassembling staff are undertaken by 5 randomly selected students. Since none of them have professional training, they believe that their disassembly proficiency is basically the same, and in order to prevent some additional factors from causing errors in disassembly time. The impact of these bolts was 4 groups of disassembly per person, 6 bolts are disassembled per group, a total of 24 disassembly, and their average value was taken as the standard disassembly time under the conditions specified for the bolt. According to the above disassembly methods, the disassembly time of the 30 disassembly schemes of the bolt was calculated, and the statistical results are shown in Table 1.

As can be seen from Table 1, the factors affecting the disassembly process are qualitative factors. Due to quantitative factors, in order to better compare the impact of each factor on the disassembly time, this article uses a combination of direct observation and comparative analysis and gray correlation analysis from disassembly data. The analysis method is used to compare the primary and secondary effects of various factors. In order to facilitate the comparative analysis of the factors in the disassembly process, the calculation principle of the gray correlation is introduced as follows.

Based on the results in Table 1, it can be seen that the factors that affect the analysis of bolted disassembly time

Table 1. Disassembly experiment results.

Data Code	Tightening Torque	Tool Code	Pull Force	Time
1	25	I	28	18.07
2	50	I	28	18.12
3	75	I	28	22.34
4	25	II	28	29.56
5	50	II	28	30.49
6	75	II	28	34.37
7	25	I	24	18.06
8	50	I	24	18.56
9	75	I	24	22.38
10	25	II	24	30.77
11	50	II	24	31.02
12	75	II	24	35.73
13	25	I	20	18.25
14	50	I	20	18.94
15	75	I	20	23.17
16	25	II	20	29.18
17	50	II	20	30.86
18	75	II	20	27.04
19	25	I	15	18.21
20	50	I	15	18.80
21	75	I	15	23.65
22	25	II	15	33.41
23	50	II	15	34.71
24	75	II	15	39.92
25	25	I	12	19.05
26	50	I	12	20.17
27	75	I	12	24.63
28	25	II	12	34.05
29	50	II	12	34.87
30	75	II	12	40.09

are both qualitative factors, such as the type of disassembly tool, and quantitative factors, such as the tightening torque of the bolt and the pulling force of the dismantling staff. The analysis of the factors affecting time becomes more difficult, so the combined analysis method of direct observation and comparative analysis and gray correlation analysis is used for the disassembly data described above. With the data results, it can be

easily seen that when the tools are different, the data changes greatly, indicating that the impact is greatest.

The changes in the tightening torque and the strength of the disassembly workers are not easy to distinguish. In order to quantify the impact of these two factors, and to analyze the primary and secondary factors of the two factors, the gray of disassembly tools I and II The association analysis process is demonstrated as follows:

(1) The gray correlation analysis for tool I is as follows:

The results of 15 sets of data of the test of the disassembly tool I with the results of Table 1, combined with the basic idea of gray correlation analysis, can be obtained from the analysis of the parent factor; that is, the disassembly time data is:

$$\begin{aligned} y_0 &= (y_1, y_2, n, y_n) \\ &= \begin{pmatrix} 18.07, 18.12, 22.34, 18.06, 18.56, 22.38, 18.25, \\ 18.94, 23.17, 18.21, 18.80, 23.65, 19.05, 20.17, \\ 24.63 \end{pmatrix}. \end{aligned}$$

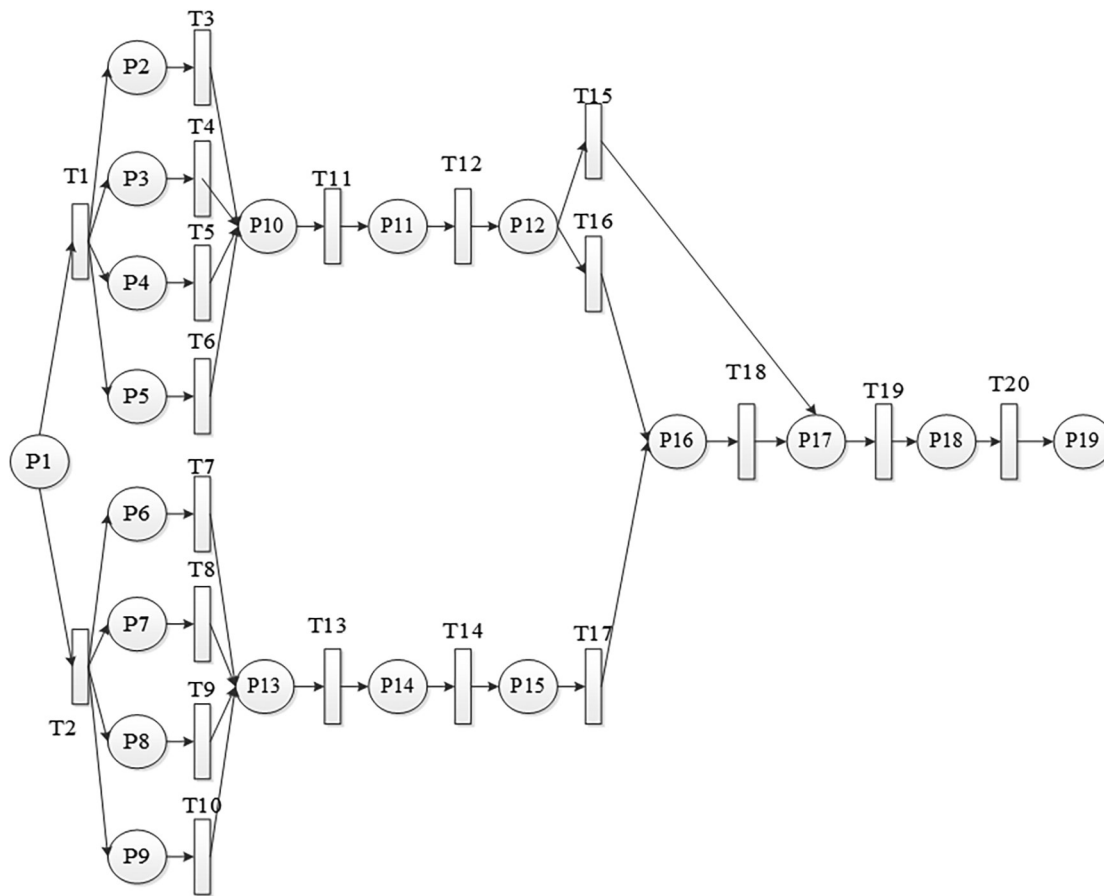


Figure 3. Automotive transmission expanded stochastic Petri net.

Construction of extended stochastic petri nets model

Based on the basic theory of Petri nets modeling method, combined with the composition of typical automobile transmission device, referring to the disassembly priority matrix shown in formula (4), dredging the constraints between parts and disassembly sequence, an extended stochastic Petri nets model of automobile transmission device is constructed, as shown in Figure 3. Among them, the places represent the parts to be removed, and the transitions represent the removal process and the constraint relationship. The label 2 in Figure 5.1 shows the four bolts on the left end cap. For convenience, the four bolts are represented by bolt I-1, bolt I-2, bolt I-3, and bolt I-4, respectively. The same is true for bolt 10.

The place in the Petri network represents the spare parts of the used products. The specific meaning of each place is shown in Table 1. The transition in the Petri net represents the disassembly process of the used product. Due to the uncertainty of the disassembly environment, when assigning the disassembly time, cost, and energy for the disassembly process of each part, this article fits the experimental data to obtain the corresponding random variables according to actual work experience. The

meaning of each transition is shown in Table 2, where $N(\mu, \sigma^2)$ is a normal distribution, $U(a, b)$ is a uniform distribution, and $\text{Logn}(\log(\mu), \sigma^2)$ is a lognormal distribution.

Incidence matrix

According to the actual characteristics of automotive transmission device and the construction method of disassembly priority matrix in disassembly Petri nets theory, the Petri nets connection matrix of automotive transmission device shown in Figure 2 is obtained as shown in the following equation:

Table 2. Meaning of each place.

Place	Meaning	Place	Meaning
P1	Start	P11	Bearing I
P2	Bolt I-1	P12	Axis
P3	Bolt I-2	P13	Right End Cap
P4	Bolt I-3	P14	Bearing II
P5	Bolt I-4	P15	Bushing
P6	Bolt II-1	P16	Gear
P7	Bolt II-2	P17	Key
P8	Bolt II-3	P18	Cover
P9	Bolt II-4	P19	Finished
P10	Left End Cap		

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (5)$$

Because of the particularity of the transmission device, it is considered in this article

that when disassembling the parts on the transmission device of an automobile, all the parts on it can be disassembled only by disassembling along the horizontal direction.

Disassembly state identification vector

Each disassembly state of the product can be represented by the corresponding disassembly state identification vector. The initial disassembly state identification vector is set to M_0 , and the final disassembly state identification vector of the product is set to M after the disassembly task is completed. Each disassembly operation must not violate the disassembly constraints between parts.

According to the definition of disassembly state identification vector, the initial disassembly state identification vector M_0 of the product to be disassembled is as follows:

$$\vec{M}_0 = [1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]^T.$$

If the object of disassembly is the part 17-case cover in Figure 2, the final disassembly state identification vector M of the product is as follows:

$$\vec{M} = [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1]^T.$$

Solution method for model

In this article, the object of disassembly is the part 17-case cover-in the automobile transmission device shown in Figure 2. In order to get part 17 by disassembly,

relevant parameters are set according to the relevant constraints of minimum expected disassembly fee model, maximum disassembly degree model, minimum expected cost model and the theory of multiobjective genetic algorithm, and the optimal solutions are obtained respectively.

Genetic algorithm is a kind of mimicking a series of changes in the process of biological evolution, such as chromosome replication, crossover, mutation, etc., and abstracts the behavior of this chromosome into a mathematical model, which is applied to practical problems.

When dismantling a waste product with a relatively simple structure, it is generally possible to determine a feasible disassembly sequence based on actual experience, and subjectively determine the optimal disassembly sequence. However, it is difficult to subjectively find a feasible disassembly sequence for dismantling disused structures with complex structures. Therefore, it is necessary to construct a disassembly network graph model to find a feasible disassembly sequence. However, when there are many paths for disassembling the target parts, it is difficult to obtain a specific optimal disassembly sequence.

In order to solve the multiobjective stochastic comprehensive optimization model with the lowest cost, the shortest time, and the least energy, the thesis designs a multiobjective optimization algorithm based on genetic algorithm to solve the optimal disassembly sequence.

Definition of the parameters

The main consideration is to construct a comprehensive optimization model with the shortest disassembly time, the lowest disassembly cost, and the lowest disassembly energy under the premise of meeting the disassembly constraints.

Minimum expected disassembly time model

In a given random disassembly network diagram, if there is a disassembly path from the disassembly starting point to the target disassembly node, for any other disassembly path, the following inequality holds, namely:

$$E[T(t, x)] \leq E[T(t, x')].$$

The path is called the minimum expected value of the disassembly of the product from the disassembly start point to the target disassembly end point, where there is the expected value of the disassembly time required for the target disassembly along the path, that is, the average disassembly required for the target part disassembly along the path. The model that solves the minimum

expected value path is called the disassembled minimum expected value model.

The objective function for disassembling the minimum expected value model is:

$$\min E\left[\sum_{i,j} t_{ij}x_{ij}\right].$$

The constraints it meets are as follows:

$$\begin{cases} x_{ij} \in \{0, 1\} \\ i, j \in V \\ (i, j) \in E \end{cases}.$$

Maximum disassembly model

In a given random disassembly network diagram, if there is a disassembly path from the disassembly starting point to the target disassembly node, for any other disassembly path, the following inequality holds:

$$P\{T(t, x) \leq T'\} \geq P\{T(t, x') \leq T'\}.$$

The path is said to be the maximum disassembly degree path from the disassembly start point to the target disassembly node. The model that solves the maximum disassembly degree path is called the maximum disassembly degree model. Among them are the disassembly time of a given product, and the disassembly path and the disassembly degree at a given time, respectively.

Therefore, the objective function of the maximum disassembly model of the disassembly problem can be expressed as follows:

$$\max PT(t, x) \leq T'.$$

The constraints it meets are as follows:

$$\begin{cases} x_{ij} \in \{0, 1\} \\ i, j \in V \\ (i, j) \in E \end{cases}.$$

D-Minimum disassembly time model

In a given random disassembly network diagram, if there is a disassembly path from the disassembly starting point to the target disassembly node, for any other disassembly path, the following inequality holds:

$$\begin{aligned} \min\{\bar{T} \mid P\{T(t, x) \leq \bar{T}\} \geq D\} \\ \leq \min\{\bar{T} \mid P\{T(t, x) \leq \bar{T}\} \geq D\}. \end{aligned}$$

The path is called the D-minimum time model that is disassembled from the disassembly starting point to the

target disassembly node. Wherein, the disassembly time of the feasible disassembly path at the disassembly degree D is given. The model that solves the D-shortest disassembly time path is called the D shortest disassembly time model.

For the convenience of solving, the mathematical expression of the D-shortest disassembly time model is now given, which is as follows. According to the related concepts of the opportunity constraint programming model, the objective function of the model can be expressed as

$$\min \bar{T}.$$

The constraints it meets are as follows:

$$\begin{cases} P\{T(t, x) \leq \bar{T}\} \geq D \\ x_{ij} \in \{0, 1\} \\ i, j \in V \\ (i, j) \in E \end{cases}.$$

The disassembly energy can be expressed by power because of the disassembly of the old vehicle transmission. At the same time, according to the principle of stochastic simulation algorithm and the constraints of planning model, the parameters of the algorithm are calibrated as follows:

Times of stochastic simulation $N = 300,000$;

Initial state identification vector: $M_0 = [1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]^T$;

The target disassembly status identification vector is set as $M = [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]^T$;

Given disassembly time in maximum disassembly degree model $t = 220s$, disassembly fee $s = \text{CNY}40$, disassembly energy $w = 0.085 \text{ kW}\cdot\text{h}$;

Given probability in minimum cost disassembly model $p_1 = p_2 = p_3 = 95\%$;

The correlation matrix A is shown in formula (5).

Using the multiobjective disassembly optimization algorithm based on stochastic simulation, the three programming models proposed in this article are solved, and the results are listed in the next section.

Solutions

According to the constraint equation, $\vec{A}\vec{u}_k = \vec{M} - \vec{M}_0$, the decision vector U is solved as follows:

$$U_1 = [1, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 1]^T;$$

$$U_2 = [1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 1]^T;$$

$$U_3 = [0, 1, 0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1]^T.$$

The disassembly sequences corresponding to the above three decision vectors are as follows:

Sequence 1: P1-><P2, P3, P4, P5>->P10->P11->P12->P17->P18->P19;

Sequence 2: P1-><P2, P3, P4, P5>->P10->P11->P12->P16->P17->P18->P19;

Sequence 3: P1-><P6, P7, P8, P9>->P13->P14->P15->P16->P17->P18->P19.

The probability density curves of disassembly time corresponding to disassembly Sequences 1, 2, and 3 are shown in Figure 4, respectively.

The disassembly fee probability density curves corresponding to disassembly sequences 1, 2, and 3 are shown in Figure 5.

The disassembly energy probability density curves corresponding to disassembly sequences 1, 2, and 3 are shown in Figure 6.

According to the objective function and constraints of the model at the minimum expected cost, the Pareto

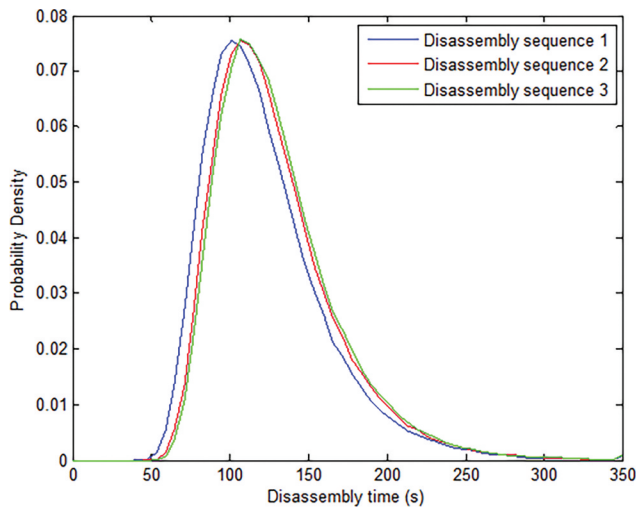


Figure 4. Probability density curves for the disassembly times.

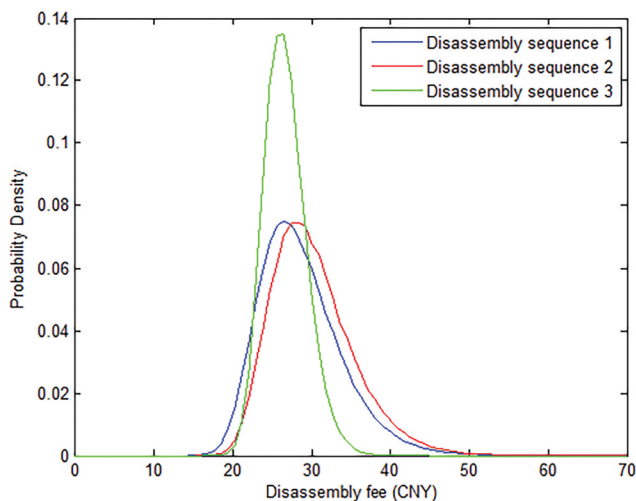


Figure 5. Probability density curves for the disassembly fees.

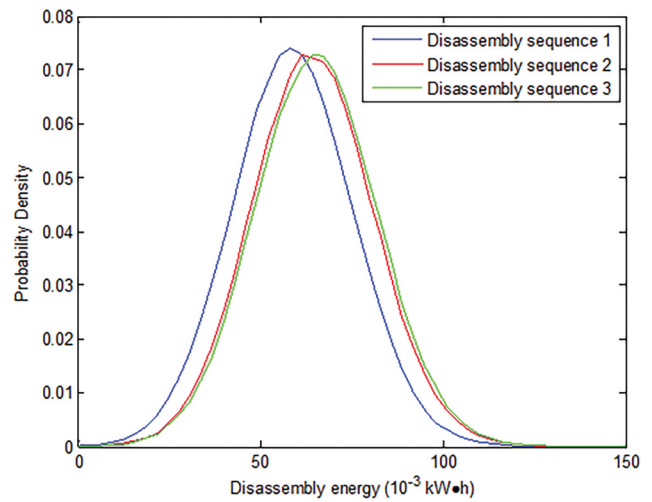


Figure 6. Probability density curve of the disassembly energy.

solution set of the disassembly sequences of the vehicle transmission device shown in Figure 2 can be obtained as shown in Table 4.

According to the objective function and maximum disassembly degree model constraints and given the disassembly times, $t_1 = 220$ s and $t_2 = 200$ s, disassembly fees, $s_1 = ¥ 40$ and $s_2 = ¥ 35$, and disassembly energy usages, $w_1 = 0.090$ kW·h and $w_2 = 0.085$ kW·h, the probabilities that the disassembly task can be completed at a given cost are shown in Table 5.

According to the objective function, maximum disassembly degree model constraints, and given the disassembly times, disassembly fees and disassembly energy usages, the probabilities are $p_1 = p_2 = p_3 = 95\%$ and $p_{1'} = p_{2'} = p_{3'} = 90\%$. The minimum costs of the three feasible disassembly sequences given the disassembly probabilities are shown in Table 6.

Conclusion

The results of Tables 3, 4, and 5 show that the objective functions of disassembly sequences 1, 2, and 3 do not dominate each other. Therefore, the three disassembly sequences are Pareto solutions of the minimum expected disassembly fee model, the maximum disassembly degree model, and the minimum cost model proposed in this article.

When the decision-makers have different preferences, different disassembly optimization results will be obtained when solving the disassembly sequence of waste products. In this article, the minimum disassembly fee model, maximum disassembly degree model, and minimum disassembly fee model are proposed so that decision makers can choose different decision schemes

Table 3. Meaning of each transition.

Transition	Meaning	Removal Time (s)	Removal Fees (¥)	Removal Energy ($10^{-3}\text{kW}\cdot\text{h}$)
T1	Ready	N (3,1)	N (1,0.2)	U (0.4,0.6)
T2	Ready	N (3,1)	N (1,0.2)	U (0.4,0.6)
T3	Remove bolt I-1	Logn(log (15),0.5)	Logn(log (4),0.3)	N (8,4)
T4	Remove boltII-2	Logn(log (15),0.5)	Logn(log (4),0.3)	N (8,4)
T5	Remove boltII-3	Logn(log (15),0.5)	Logn(log (4),0.3)	N (8,4)
T6	Remove boltII-4	Logn(log (15),0.5)	Logn(log (4),0.3)	N (8,4)
T7	Remove boltII-1	Logn(log (15),0.5)	Logn(log (3),0.2)	N (8,4)
T8	Remove boltII-2	Logn(log (15),0.5)	Logn(log (3),0.2)	N (8,4)
T9	Remove boltII-3	Logn(log (15),0.5)	Logn(log (3),0.2)	N (8,4)
T10	Remove boltII-4	Logn(log (15),0.5)	Logn(log (3),0.2)	N (8,4)
T11	Remove left End Cap	U (4,6)	U (0.8,1.0)	Logn(log(3.5),0.8)
T12	Remove bearing I	Logn(log (20),0.6)	Logn(log (4),0.2)	N (10.5,0.64)
T13	Remov right End Cap	U (4,6)	U (0.8,1.0)	Logn(log(3.5),0.8)
T14	Remove bearing II	Logn(log (20),0.6)	Logn(log (4),0.2)	N (10.5,0.64)
T15	Remove axis	N (10,0.8)	N (2,0.3)	U (3,6)
T16	Remove axis	N (10,0.8)	N (2,0.3)	U (2,6)
T17	Remove bushing	N (12,0.7)	N (3,0.2)	N (5,2)
T18	Remove gear	N (6,0.8)	N (1.5,0.2)	U (4,8)
T19	Remove key	N (8,1)	N (3,0.4)	N (4,1)
T20	Remove cover	N (5,0.5)	N (1,0.1)	U (2,3)

Table 4. Solution of the minimum expected cost disassembly model.

Sequence No.	Expected Disassembly Time (s)	Expected Disassembly Fee (CNY)	Expected Disassembly Energy ($10^{-3}\text{kW}\cdot\text{h}$)
Sequence 1	117.4864	33.6538	61.4924
Sequence 2	123.4864	35.1538	59.9926
Sequence 3	125.4864	36.1538	56.9926

Table 5. Solution of the maximum disassembly degree model.

Sequence No.	Limited Disassembly Time: 220 s	Limited Disassembly Fee: CNY40	Limited Disassembly Energy: 0.085 $\text{kW}\cdot\text{h}$
Sequence 1	97.4287%	96.8190%	92.1220%
Sequence 2	96.9820%	95.1833%	93.4010%
Sequence 3	96.7790%	93.8940%	95.3320%

Table 6. Solution of the minimum disassembly cost model.

Sequence No.	Minimum Disassembly Time (s)	Minimum Disassembly Fee (CNY)	Minimum Disassembly Energy ($10^{-3}\text{kW}\cdot\text{h}$)
Sequence 1	197.6627	38.2920	88.9595
Sequence 2	203.1698	39.7808	87.3564
Sequence 3	205.2454	40.7971	84.5262

according to market demand and internal needs of enterprises. More intelligent methods is further to test the proposed models.

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